

Department of Mathematics and Applied Mathematics
3MS Metric Spaces

Tutorial 12

2024

1. Let X be an infinite set and d_D the discrete metric on it. Show that (X, d_D) is complete and bounded, but not compact.
2. Suppose K is a compact subset of a metric space (X, d) and $x \in X$ but $x \notin K$. Show that there exist disjoint, open subsets G_1 and G_2 of (X, d) such that $x \in G_1$ and $K \subseteq G_2$. (Hint: Use the version of compactness we called “having a compact topology.” You will also need the Hausdorff property.)
3. **Lemma** Every subspace of a totally bounded metric space is totally bounded.

Proof Suppose X is totally bounded and $A \subseteq X$. (1)

Let $\epsilon > 0$ be given. (2)

By assumption, there exists a finite $F \subseteq X$ such that $\bigcup\{B_X(t; \frac{\epsilon}{2}) \mid t \in F\} = X$. (3)

Let $K = \{x \in F \mid B_X(x; \frac{\epsilon}{2}) \cap A \neq \emptyset\}$. (4)

Note that K is finite. (5)

For each $x \in K$, choose $a_x \in B_X(x; \frac{\epsilon}{2}) \cap A$. (6)

Note that $d(a_x, x) < \frac{\epsilon}{2}$ for all $x \in K$. (7)

We show that $\bigcup\{B_A(a_x; \epsilon) \mid x \in K\} = A$: (8)

Take $a \in A$. (9)

Then $a \in B_X(y; \frac{\epsilon}{2})$ for some $y \in F$. (10)

So $a \in B_X(y; \frac{\epsilon}{2}) \cap A$. (11)

So $y \in K$. (12)

Now $d(a, a_y) \leq d(a, y) + d(y, a_y) < \frac{\epsilon}{2} + \frac{\epsilon}{2}$. (13)

So $a \in B_A(a_y; \epsilon)$ and $y \in K$, as desired. (14)

- (a) How do we know the set K mentioned in line (5) is finite?
- (b) How do we know the set K mentioned in line (5) is non-empty?
- (c) The argument from line (9) to line (14) shows that $\bigcup\{B_A(a_x; \epsilon) \mid x \in K\} \supseteq A$. Why is the reverse inclusion clear?
- (d) How do we know the y mentioned in line (10) exists?

4. (a) Let (X, d) be a metric space, $c \in X$ and (x_n) a sequence in X . Prove that if, for all $\epsilon > 0$, we have $d(x_n, c) < \epsilon$ for infinitely many n , then (x_n) has a subsequence converging to c .
 (b) State the contrapositive of this statement. (The contrapositive of $P \implies Q$ is $\neg Q \implies \neg P$. A statement and its contrapositive are logically equivalent.) This is useful in proving one implication of the equivalent statements in Theorem 9.22 of your notes.)
5. Suppose that A and B are disconnected subsets of a metric space X . Which of the following are necessarily disconnected? Give an appropriate proof or counter-example.
 - (a) $A \cup B$
 - (b) A'
 - (c) $A \cap B$
6. Suppose that A and B are connected subsets of a metric space X . Which of the following are necessarily connected? Give an appropriate proof or counter-example. (You will be able to re-use one of your counter-examples above; which one?)
 - (a) $A \cup B$
 - (b) A'
 - (c) $A \cap B$
7. Find the error in the argument below, and then find a counter-example to show the “result” false.

Incorrect statement If $f : X \rightarrow Y$ is a continuous function between metric spaces and X is connected, then Y is connected.

Incorrect argument If Y is disconnected, there exists $g : Y \rightarrow \mathbf{2}$, non-constant and continuous. Then $g \circ f : X \rightarrow Y \rightarrow \mathbf{2}$ is non-constant and continuous, so X is disconnected.